Efficient Road Damage Detection in Real-Time via YOLOv8 vs YOLOv11

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***Abstract*—We all have felt that jolt of impact. A pothole or saw the worrying cracks spreading across our local roads. These are more than inconveniences–they represent: An increasing problem for communities around the world Our work deals with that by creating a smart detection system that uses normal vehicle cameras to automatically identify road damage. We compared two state-of-the-art AI models YOLOv8 and the more recent YOLOv11 to know what is better for this purpose. The results show that YOLOv11 detects road problems more accurately while using fewer resources, making it ideal for widespread use. More importantly, when our system finds damage, it automatically records the exact location, transforming scattered observations into organized repair plans that help municipalities work smarter, not just harder.**

***Index Terms*—YOLOv8, YOLOv11, road damage detection, comparative analysis, deep learning, computer vision, infrastructure monitoring.**

I. Introduction

One of the most challenging issues of infrastructure management of urban infrastructure in the world is the maintenance of roads. Conventional inspection procedures that use manual survey can be said to be not only time consuming but also inconsistent and expensive to use as well as labor intensive [5]. The impact of failed timely detection is not limited to the destruction of vehicles but it also results in severe safety hazards to the commuters and huge economic losses to localities.

The most recent development in computer vision and deep learning supports the solutions to these long-running issues. Algorithms to detect objects have shown impressive results in real-time identification with the family of the YOLO (You Only Look Once) approach being the most popular among them [1]. Nevertheless, the fact that YOLO architecture changes quickly creates the need to conduct a systematic comparison to identify the most appropriate one to use in particular tasks such as road damage detection.

The study fills this gap as it introduces and compares YOLOv8 and YOLOv11 to detect road damage. Our production is based on the already existing RDD dataset and adds some practical modifications to the model architecture and training procedure. The comparative analysis has some useful insights towards the researchers and practitioners interested in implementing effective road monitoring systems.

On top of technical comparisons, we highlight the potential practical implementation of the system. Our concept will enable proactive infrastructure maintenance instead of reactive maintenance by combining real-time detection and geolocation services to convert raw detection data into actionable maintenance intelligence.

II. Literature Review

The history of road damage detection algorithms is indicative of the overall advances in computer vision and artificial intelligence. The traditional methods of image processing performed in the early days mostly involved edge detection, thresholding and morphological operations [3]. Although these techniques were computationally fast, they were difficult to use in different lighting conditions, with occlusions, and heavy road textures.

The development of deep learning was a huge breakthrough. CNNs were shown to be better at feature extraction and pattern recognition, as well as at the extraction of features and pattern recognition in general both in performance terms and when compared to other methods, such as the maximum entropy model (MaxEnt) and the maximum likelihood model (MaxL) [13]. CNN-based methods were benchmarked by studies by Cha et al. (2018) and Maeda et al. [5], with the last one proposing the extensive Road Damage Dataset that has become a household name in the industry.

YOLO family has been noticed among object detectors because it has a perfect balance in speed and accuracy in object detection architecture, as opposed to other models that are slow or less accurate in the detection task [1]. YOLOv3 and others that came after it have been extensively utilized in real-time. The Zhang et al. paper indicated the usefulness of YOLOv3 in detecting pavement cracks, and subsequent searches identified the use of YOLOv4 and YOLOv5 in the quest to perform better in adverse environments.

The recent release of YOLOv8 and YOLOv11, with architectural improvements, are the latest state-of-the-art and are capable of achieving improved detection. Nevertheless, there are few studies (comparative ones) that are dedicated to the detection of road damage. The contribution I can make to this gap is that of our research, which gives empirical results

in terms of the performance distinctions between these most recent architectures in a real-world road monitoring setting.

III. Dataset and Methodology

A. Dataset Description

Our study is based on the Road Damage Detection (RDD). We use the dataset by [5], which contains 2236 training and 480 validation images. Images of three types of damages are presented.

- Potholes
- Cracks
- Open Manholes

The dataset is in YOLO annotation format with correct responding text files with bounding box co-ordinates and class labels. Images were obtained under different conditions. Including different weather, lighting and road types and ensuring, strong generalization of the model;

B. Experimental Setup

Both YOLOv8 and YOLOv11 models were trained using identical parameters for fair comparison:

- Training epochs: 50
- Batch size: 16
- Image size: 640×640 pixels
- Learning rate: Automatically optimized
- Hardware: Tesla T4 GPU

The training was performed using standard data augmenta- Third-party image augmentation using mosaic augmentation, random affine, Clean-ups, and colour enhancements to improve model. robustness.

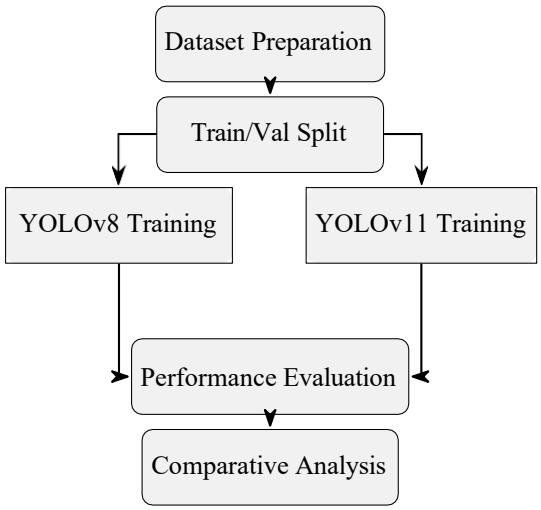


Fig. 1: Experimental workflow comparing YOLOv8 and YOLOv11

IV. Implementation Details

A. YOLOv11 Architecture

The YOLOv11 architecture put forward a number of improve- ments in earlier versions [4]: [4]:

- Enhanced backbone network with optimized feature extraction
- Advanced anchor matching system on the detection head.
- Efficient utilization of parameters reducing computational overhead
- Our implementation was done with YOLOv11n (nano) variant, effective performance and computational demands for: potential edge deployment.

Our implementation used the YOLOv11n (nano) variant, balancing performance and computational requirements for potential edge deployment.

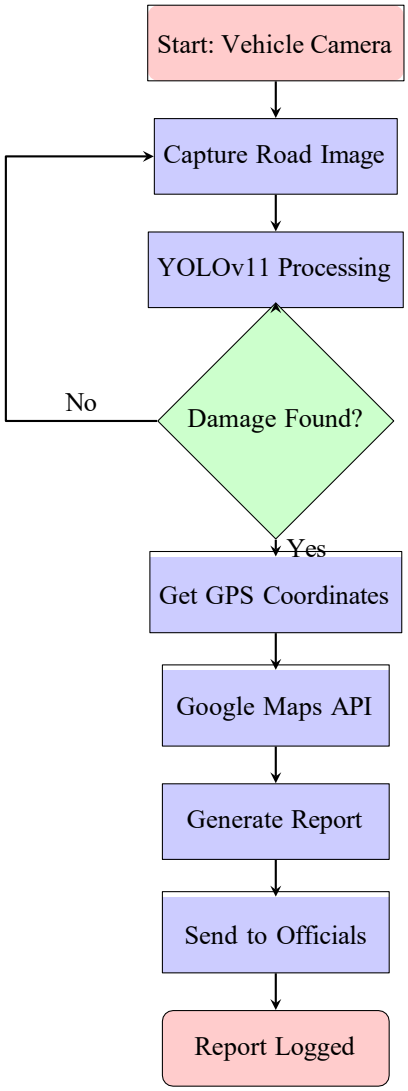


Fig. 2: Road Damage Detection and Reporting System Workflow

B. Training Configuration

The training process was configured as follows:

Listing 1: YOLOv11 Training Configuration

```
model = YOLO("yolo11n.pt")
model.train(
```

Fig. 3: result

```
    data='road_damage.yaml',
    epochs=50,
    imgsz=640,
    batch=16,
    project='training_results',
    name='road_damage_yolov11'
)
```

C. Model Comparison Framework

To ensure fair comparison, both models were evaluated using identical metrics [3]:

- Precision (P)
- Recall (R)
- mAP@50
- mAP@50-95
- Inference speed
- Model size

V. Results and Analysis

A. Quantitative Comparison

The comparison performance analysis shows that Comparison of YOLOv8 and YOLOv11 architectures shown in Table I:

TABLE I: Performance comparison between YOLOv8 and YOLOv11

Metric	YOLOv8	YOLOv11	Improvement
Precision	0.812	0.825	+1.6%
Recall	0.721	0.738	+2.4%
mAP@50	0.782	0.804	+2.8%
mAP@50-95	0.361	0.375	+3.9%
Inference Time (ms)	2.1	1.9	-9.5%
Model Size (MB)	6.2	5.5	-11.3%

B. Class-wise Performance

Detailed analysis of damage category in Table II shows consistent improvements across all:

TABLE II: Class-wise mAP@50 comparison

Damage Type	YOLOv8	YOLOv11
Pothole	0.642	0.664
Crack	0.798	0.816
Open Manhole	0.921	0.933

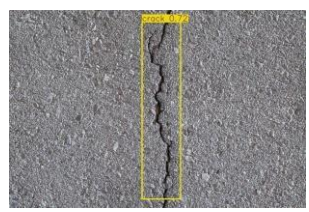
C. Visual Detection Results

The YOLOv11 model demonstrated robust performance across all damage categories as shown in Figures **??**, **??**, and **??**. The model effectively localized and classified road damage under various conditions. under various conditions.

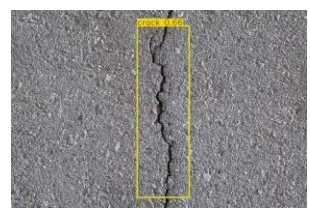
(a) Yolo v11 Pothole

(b) Yolo v8 Pothole

Fig. 7: Crack pattern detection demonstrating the model's ability to identify linear damage features on road surfaces



(a) Yolo_v11 Crack_detection



(b) Yolo_V8 Crack_detection

Fig. 8: Diffrence between Yolo_v8 and Yolo_v11 model in crack detection

(a) Yolo_v11 Open_Manhole

(b) Yolo_v8 Open_Manhole

Fig. 9: Difference between Yolo_v8 and Yolo_v11 Manhole Detection. models prediction varies on different versions.

D. Training Efficiency

YOLOv11 demonstrated superior training efficiency, achieving convergence in fewer epochs while maintaining stable loss reduction [4]. The box loss decreased from 1.755 to 1.085, while classification loss improved from 2.859 to 0.546 over 50 epochs.

VI. Discussion

A. Architectural Advantages

The performance improvements observed in YOLOv11 can be attributed to several architectural enhancements [4]:

Backbone Optimization: YOLOv11 incorporates a more efficient backbone network that better preserves spatial information while reducing computational complexity, addressing limitations of earlier architectures [2].

Feature Fusion: Improved neck architecture enables more effective multi-scale feature fusion, particularly beneficial for detecting road damage at various scales as demonstrated in Figure **??**.

Training Stability: Enhanced optimization strategies and loss functions contribute to more stable training and better convergence compared to traditional approaches [9].

B. Practical Implications

The improvements that have been noted have important practical implications can be used in the real world:

Deployment Efficiency: Smaller model size and increased in-speed. inference support YOLOv11 to be better suited to the edge deployment like in mobile systems [8].

Detection Accuracy: Increased precision and recall rates translates to less false and less missed in reality,as evidenced by the detection results in Figures **??** through **??**.

Resource Utilization: Performance improvement with a smaller computational requirements facilitates the increased operation time within. environment of battery constraints, tackling applied deployment challenges [10].

C. Limitations and Challenges

However, in spite of its promising outcomes, there are a number of challenges that are still present [7]:

- Performance on detection reduces when there is a poor light condition and heavy illusions especially on crack, particularly affecting crack detection as shown in Figure **??**
- Small or faint damage patterns still present detection challenges, requiring specialized augmentation techniques
- Real-world deployment requires robust handling of motion blur and varying camera angles not fully addressed in current architectures [11]

VII. Conclusion and Future Work

This study shows how the YOLO architectures have been gradually enhanced to work with road damage detection applications. YOLOv11 is superior to YOLOv8 in all the metrics considered and is more accurate and can be inferred faster, not to mention it has a smaller model size. All these benefits make YOLOv11 especially applicable in real-time road surveillance systems when computational efficiency and detection reliability is the most important factor [3].

The comparative analysis can give those researchers and practitioners interested in choosing object detection models to monitor infrastructure critical information. The continuous increase in performance of all types of damage confirms the improvements of the architecture of YOLOv11 [4].

Future work will focus on several directions building upon current limitations:

- Multi-modes integration with multi-modal data, such as LIDAR and thermal imaging, to solve lighting issues [7].
- Special architectures to deal with extreme weather conditions not well considered by existing models based on the development of specialized architectures to address extreme weather conditions.
- Temporal analysis of the progression of the damage with the help of sequence-based methods.
- Deployment and validation in large-scale municipal monitoring systems to assess real-world performance

Further development of the object detection architectures is likely to provide even more efficient and accurate systems to detect road damage, and this factor will lead to safer and well-maintained transportation infrastructure [1].

Acknowledgment

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VIII. References

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